AyuAware Learn Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Learn section is the **Ayurvedic Wisdom Academy** — a structured educational experience where users build deep understanding of Ayurvedic principles. It answers the fundamental question: "*How can I truly understand Ayurveda so I can apply it to my own health?*"

This isn't a random content library with wellness articles. It's a curriculum-based learning system that:

Dimension	What We Offer	Why It Matters
Personalized Learning	Content surfaced based on constitution, imbalances, and active projects	Generic content doesn't connect to user's actual health journey
Twin as	Digital Twin guides learning, explains	Learning is better with a
Structured	Organized curriculum with clear	Random lessons don't build deep
Learn in	Concepts explained when encountered	Knowledge is best absorbed when
Theory +	Every concept includes practical	Understanding without application is
Meaningful Progression	Levels unlock deeper content and capabilities	Progress should feel rewarding and enable new possibilities

Why It Matters

Most health apps either have no educational content or dump articles without structure. Users encounter terms like "Vata," "Agni," or "Ama" without understanding what they mean or why they matter for their health.

Ayurveda is a comprehensive system with thousands of years of wisdom. To truly benefit from it, users need to understand the principles — not just follow instructions blindly. When someone understands WHY they should eat warm foods (to support Agni) or WHY they should have consistent routines (to balance Vata), they're more likely to follow through and make lasting changes.

Key Differentiator

Generic wellness app: "Article: 10 Benefits of Turmeric. Next: Random article about sleep."

AyuAware Learn: "You're 2 lessons away from Level 4! Since you're working on your Nidra Protocol, I recommend learning about 'Sleep & the Doshas' next — it'll help you understand WHY the practices work. This lesson includes a personalized section showing how YOUR Pitta-Vata constitution affects YOUR sleep patterns."

Structured. Personalized. Connected to health journey. Twin-guided.

2. Section Overview

2.1 Information Architecture

```
Learn Section
│
├── 🎓 Dashboard (Main View)
│    ├── Twin Learning Guide
│    ├── Continue Learning (in-progress lesson)
│    ├── Your Learning Paths
│    ├── Recommended For You
│    ├── Topics Mastered
│    ├── Quick Stats
│    └── Concept Library Quick Access
│
├── 📚 Learning Paths
│    ├── Foundation Path (Required)
│    ├── Nutrition & Digestion Path
│    ├── Lifestyle & Routine Path
│    ├── Sleep & Recovery Path
│    ├── Mind & Emotions Path
│    ├── Seasonal Living Path
│    ├── Advanced Practices Path (Level 4+)
│    ├── Dosha Deep Dive Path (Level 4+)
│    └── Herbs & Substances Path (Level 5+)
```

```
│
├── 📖 Lesson View
│   ├── Content Area (text, images, video, interactive)
│   ├── Personalized Context ("For Your Constitution")
│   ├── Progress Indicator
│   ├── Knowledge Checks (quizzes)
│   ├── Key Takeaways
│   └── Practical Application
│
├── 📚 Concept Library (Encyclopedia)
│   ├── Core Concepts (Doshas, Gunas, etc.)
│   ├── Body Systems (Dhatus, Srotas, etc.)
│   ├── Health Markers (Agni, Ama, Ojas, etc.)
│   ├── Practices (Dinacharya, Ritucharya, etc.)
│   ├── Treatments (Panchakarma, Rasayana, etc.)
│   └── Substances (Herbs, Foods, etc.)
│
├── 📊 Progress & Achievements
│   ├── Level Progress
│   ├── Paths Completed
│   ├── Statistics
│   ├── Achievements/Badges
│   └── Learning History
│
└── 🔍 Learn in Context (App-wide)
    └── Quick concept cards when tapping terms
```

2.2 The Level System

Meaningful progression that unlocks capabilities:

Level	Name	XP Required	Lessons	Unlocks
1	Newcomer	0	0	Foundation Path, Basic features
2	Student	100	~3	Intermediate paths, Concept Library
3	Apprentice	300	~8	All intermediate paths, Deeper content
4	Practitioner	600	~15	Advanced paths, Protocol customization
5	Adept	1000	~25	Herbs path, Advanced assessments
6	Scholar	1500	~35	Full library access, Practitioner features
7	Master	2500	~50	Complete mastery, All features

XP Sources:

- Completing a lesson: 30-50 XP (based on length)

- Perfect quiz score: +10 XP bonus
- Completing a path: 100 XP bonus
- Learning streak (daily): 5 XP/day
- First lesson of day: 10 XP bonus

2.3 Learning Paths Overview

Foundation Path (Required - 7 Lessons):

#	Lesson	Duration	Content Focus
1	What is Ayurveda?	12 min	History, philosophy, principles
2	The Three Doshas	15 min	Vata, Pitta, Kapha basics
3	Discovering Your Prakriti	18 min	Constitutional assessment explained
4	Understanding Vikriti	15 min	Current state vs. baseline
5	The 20 Gunas (Qualities)	12 min	Heavy/light, hot/cold, etc.
6	Like Increases Like	10 min	Core Ayurvedic principle
7	Your Digital Twin & Ayurveda	10 min	How the app applies these concepts

Nutrition & Digestion Path (8 Lessons):

#	Lesson	Duration	Content Focus
1	The Six Tastes (Rasa)	15 min	Sweet, sour, salty, pungent, bitter, astringent
2	Food Combining Rules	12 min	What to eat together, what to avoid
3	Eating for Your Dosha	18 min	Constitutional dietary guidelines
4	Understanding Agni	15 min	Digestive fire types and health
5	Signs of Healthy Digestion	12 min	How to assess your digestion
6	Ama: Toxins & How to Clear Them	15 min	Toxic buildup and remedies
7	Seasonal Eating	12 min	Adapting diet to seasons
8	Fasting & Cleansing	15 min	Safe approaches to cleansing

Lifestyle & Routine Path (6 Lessons):

#	Lesson	Duration	Content Focus
1	Dinacharya: The Ideal Day	15 min	Daily routine overview
2	Morning Rituals Deep Dive	18 min	Tongue scraping, oil pulling, etc.
3	The Dosha Clock	12 min	Times of day and doshas
4	Self-Care Practices	15 min	Abhyanga, nasya, etc.

5	Exercise for Your Constitution	15 min	Movement matched to dosha
6	Evening Routine & Sleep Hygiene	12 min	Wind-down practices

Sleep & Recovery Path (5 Lessons):

#	Lesson	Duration	Content Focus
1	Sleep & the Doshas	15 min	How constitution affects sleep
2	The Science of Ayurvedic Sleep	12 min	Nidra principles
3	Sleep Disturbances by Type	15 min	Vata/Pitta/Kapha sleep issues
4	Creating Your Sleep Sanctuary	12 min	Environment optimization
5	Herbs & Practices for Sleep	15 min	Natural sleep support

Mind & Emotions Path (6 Lessons):

#	Lesson	Duration	Content Focus
1	The Three Gunas of Mind	15 min	Sattva, Rajas, Tamas
2	Emotions & the Doshas	15 min	How imbalances affect mood
3	Meditation in Ayurveda	18 min	Constitutional approaches
4	Managing Stress	15 min	Dosha-specific strategies
5	Building Mental Ojas	12 min	Cultivating resilience
6	The Mind-Body Connection	15 min	Psychosomatic relationships

Seasonal Living Path (6 Lessons):

#	Lesson	Duration	Content Focus
1	Ritucharya Overview	12 min	Seasonal living principles
2	The Six Seasons	18 min	Ayurvedic seasonal calendar
3	Winter Living	15 min	Hemanta/Shishira practices
4	Spring Living	15 min	Vasanta practices
5	Summer & Monsoon Living	15 min	Grishma/Varsha practices
6	Autumn Living & Transitions	15 min	Sharad practices

Advanced Practices Path (Level 4+ - 8 Lessons):

#	Lesson	Duration	Content Focus
1	Introduction to Panchakarma	18 min	The five cleansing actions
2	Preparatory Practices	15 min	Purvakarma procedures
3	Rasayana: Rejuvenation	18 min	Anti-aging and vitality
4	Vajikarana: Vitality Practices	15 min	Reproductive health
5	Marma Points	20 min	Energy points in the body
6	Advanced Pranayama	18 min	Breath practices for healing
7	Mantra & Sound Healing	15 min	Vibrational medicine
8	Working with Practitioners	12 min	When to seek professional help

2.4 Concept Library Categories

Core Concepts:

- Doshas (Vata, Pitta, Kapha)
- Prakriti (Constitution)
- Vikriti (Current State)
- Gunas (20 Qualities)
- Panchamahabhutas (5 Elements)
- Tridosha Theory

Body Systems:

- Dhatus (7 Tissues)
- Srotas (Channels)
- Malas (Waste Products)
- Marma (Vital Points)
- Chakras (Energy Centers)

Health Markers:

- Agni (Digestive Fire)
- Ama (Toxins)
- Ojas (Vitality)
- Tejas (Inner Radiance)
- Prana (Life Force)

Practices:

- Dinacharya (Daily Routine)
- Ritucharya (Seasonal Routine)
- Svasthavritta (Self-Care)
- Panchakarma (Cleansing)
- Rasayana (Rejuvenation)
- Pranayama (Breathwork)

Nutrition Concepts:

- Rasa (6 Tastes)

- Virya (Potency)
- Vipaka (Post-digestive Effect)
- Ahara (Food)
- Viruddhahara (Incompatible Foods)

Substances:

- Individual herbs (Ashwagandha, Brahmi, etc.)
- Formulations (Triphala, Chyawanprash, etc.)
- Minerals and metals
- Food categories

2.5 Design Philosophy

Personalized Learning Content is surfaced based on user's constitution, current imbalances, and active projects. Lessons include personalized sections ("For Your Constitution").

Twin as Teacher The Digital Twin guides learning, explains concepts in context, and answers questions. It's not just content delivery — it's tutoring.

Structured Paths Not random lessons — organized curriculum with clear progression. Foundation concepts come first, advanced content unlocks later.

Learn in Context When users encounter Ayurvedic terms elsewhere in the app, they can tap to learn instantly. Knowledge is absorbed when immediately relevant.

Theory + Practice Every concept includes practical application. After learning about Agni, users get a task: "Eat meals at consistent times today."

Meaningful Progression Levels unlock deeper content and app capabilities. Progress feels rewarding and enables new possibilities.

3. User Journey Breakdown

3.1 Learn Dashboard Experience

Context: User opens Learn section.

Twin Message:

"Welcome back to your learning journey! Since you're working on your Nidra Protocol, I recommend learning about 'Sleep & the Doshas' next — it'll help you understand WHY the practices work. You're 2 lessons away from Level 4!"

Learning Journey Card:

```
┌──────────────────────────────────────────────────────────────┐
│ 🎓 YOUR LEARNING JOURNEY                                     │
│ ──────────────────────────────────────────────────────────── │
│                                                              │
│            ┌──────────────────┐                              │
```

```
┌──────────────────────────────────────────────┐
│              ┌──────────────────┐            │
│              │   APPRENTICE     │            │
│              └──────────────────┘            │
│                                              │
│  Level Progress                              │
│  ▓▓▓
```

3.3 Browsing Learning Paths

Context: User views all learning paths.

Twin Message:

"Learning paths give you structured progression through Ayurvedic wisdom. I recommend starting with Foundation, then following your interests or health goals."

Path Categories:

Beginner Paths:

- Foundation Path (Required) — 7 lessons, ~2.5 hours

Intermediate Paths:

- Nutrition & Digestion Path — 8 lessons, ~3 hours
- Lifestyle & Routine Path — 6 lessons, ~2 hours
- Sleep & Recovery Path — 5 lessons, ~1.5 hours
- Mind & Emotions Path — 6 lessons, ~2 hours
- Seasonal Living Path — 6 lessons, ~2 hours

Advanced Paths (Level 4+):

- Advanced Practices Path — 8 lessons, ~2.5 hours
- Dosha Deep Dive Path — 6 lessons, ~2 hours

Expert Paths (Level 5+):

- Herbs & Substances Path — 10 lessons, ~4 hours

Path Status Indicators:

- ✅ Complete
- 🔄 Active (in progress)
- ⭐ Recommended
- 🔒 Locked (level requirement)

3.4 Taking a Lesson

Context: User starts "Understanding Agni" lesson.

Lesson Header:

```
Part 4 of Nutrition & Digestion Path
Understanding Agni
```

Lesson Content (Section by Section):

Section 1: What is Agni?

```
Agni is the Sanskrit word for "fire" and in Ayurveda
it represents all transformative processes in the
body - most importantly, your digestive fire.
```

```
Just as a fire transforms wood into ash and energy,
your Agni transforms food into nutrients, energy,
and eventually, the tissues of your body (Dhatus).

[Interactive: The Fire Metaphor - tap to explore]
```

Personalized Context Box:

```
┌──────────────────────────────────────────────────┐
│ 💡 FOR YOUR CONSTITUTION                         │
│                                                  │
│ As a Pitta-Vata type, you likely have naturally  │
│ strong but variable Agni. Your Pitta gives       │
│ digestive strength, but Vata can make it         │
│ irregular.                                       │
│                                                  │
│ This is called "Vishama Agni" - we'll cover      │
│ this in the next section.                        │
└──────────────────────────────────────────────────┘
```

Progress Indicator:

```
Progress: Section 2 of 5
[▓▓▓
```

```
✅ Correct!

Vishama Agni (variable fire) is associated with
Vata because Vata's nature is changeable and
irregular. This is why your digestion might be
strong some days and weak others.

[Continue →]
```

3.6 Lesson Completion

Context: User finishes a lesson.

Twin Message:

"🎉 Excellent! You've completed 'Understanding Agni'! This knowledge will help you understand why certain foods work better for you and why eating at regular times matters."

Completion Summary:

```
✅ LESSON COMPLETE
Understanding Agni
──────────────────────────────────────────

Time spent: 18 minutes
Quiz score: 4/5 (80%)

+50 XP earned
Progress to Level 4: 85% (+10%)

──────────────────────────────────────────

KEY TAKEAWAYS:
• Agni is your digestive & metabolic fire
• Your Agni type: Vishama (variable)
• Regular meal times help balance Agni
• Signs of healthy vs impaired Agni

──────────────────────────────────────────

PRACTICAL APPLICATION:
Based on this lesson, I've added a suggested
task to your Today view:
```

```
"Eat meals at consistent times today"
```

Path Progress: 5/8 lessons complete
62%

[Review Lesson] [Next: Signs of Healthy Digestion]

3.7 Browsing Concept Library

Context: User opens Concept Library.

Twin Message:

"The Concept Library is your Ayurvedic encyclopedia. Tap any concept to learn more, or ask me to explain anything."

Category Browser:

📂 CATEGORIES

```
🌀 CORE CONCEPTS
Doshas, Gunas, Prakriti, Vikriti,
Panchamahabhutas
```

```
🏛️ BODY SYSTEMS
Dhatus (7 Tissues), Srotas (Channels),
Malas (Waste Products), Marma Points
```

```
📊 HEALTH MARKERS
Agni, Ama, Ojas, Prana, Tejas
```

[... more categories ...]

🔤 BROWSE A-Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

3.8 Viewing a Concept

Context: User taps on "Agni" in Concept Library.

Concept Detail:

```
🔥 AGNI
Sanskrit: अग्नि
Meaning: Fire, Digestive Fire
──────────────────────────────────────

DEFINITION
Agni is the transformative principle in
Ayurveda, representing all metabolic and
digestive processes. It transforms food into
energy, nutrients, and tissues while burning
away waste and toxins.
```

Personalized Context:

```
💡 YOUR AGNI

Based on your constitution and current state,
your Agni type is likely:

Vishama Agni (Variable)

This means your digestion fluctuates -
sometimes strong, sometimes weak.
Associated with Vata elevation.

[Take Agni Assessment →]
```

Types/Subtopics:

```
📖 TYPES OF AGNI

Jatharagni (Main digestive fire)

Located in stomach and small intestine.
Responsible for initial food breakdown.

Bhutagni (5 elemental fires)

Transform food into elements the body can use.
```

```
│ One for each element.                                │
└──────────────────────────────────────────────────────┘
```

```
Dhatvagni (7 tissue fires)
```

```
┌──────────────────────────────────────────────────────┐
│ Present in each dhatu (tissue), enabling             │
│ nourishment and transformation.                      │
└──────────────────────────────────────────────────────┘
```

Agni States Table:

State	Dosha	Signs
Sama	Balanced	Perfect digestion
Vishama	Vata	Variable, irregular
Tikshna	Pitta	Sharp, intense
Manda	Kapha	Slow, sluggish

Related Concepts:

```
🔗 RELATED CONCEPTS
[Ama] [Dhatus] [Ojas] [Digestion]
[Jatharagni] [Metabolism]
```

Related Lessons:

```
📚 LEARN MORE

Related Lessons:
• Understanding Agni (Completed ✓)
• Signs of Healthy Digestion
• Strengthening Your Agni
```

3.9 Learn in Context (App-Wide Feature)

Context: User is in Health section and taps on "Vata"

Quick Concept Card (Modal/Sheet):

```
┌──────────────────────────────────────────────────────┐
│ 💨 VATA                              [Full Page →]   │
│ ────────────────────────────────────────────────────  │
│                                                      │
│ The dosha of movement and change, composed of        │
│ Air and Space elements.                              │
│                                                      │
│ Qualities: Light, dry, cold, mobile, rough           │
```

```
Your Vata Status:
Currently elevated (40% vs 25% baseline)

[Learn More] [See in My Twin]
```

This quick-reference card appears as a modal when users tap any Ayurvedic term throughout the app.

3.10 Viewing Progress & Achievements

Context: User views their learning progress.

Progress Dashboard:

```
🎓 LEVEL 3: APPRENTICE
──────────────────────────────────────

Progress to Level 4
[▓▓▓
```

```
┌─────┐ ┌─────┐ ┌─────┐ ┌─────┐
│ 🌱  │ │ 📚  │ │ 🔥  │ │ 🎯  │
│First│ │Found│ │5 Day│ │Quiz │
│Lesson│ │ation│ │Streak│ │Master│
└─────┘ └─────┘ └─────┘ └─────┘

┌─────┐ ┌─────┐ ┌─────┐ ┌─────┐
│ 🔒  │ │ 🔒  │ │ 🔒  │ │ 🔒  │
│Path │ │30 Day│ │Expert│ │Comple│
│Master│ │Streak│ │Level│ │tionist│
└─────┘ └─────┘ └─────┘ └─────┘
```

3.11 Personalized Recommendations

Context: User views "Recommended For You" section.

Recommendation Card:

```
┌──────────────────────────────────────────────┐
│ 🌙 Sleep & the Doshas                        │
│ How constitution affects your sleep          │
│ ──────────────────────────────────────────── │
│ ⏱️ 15 min  •  📊 Intermediate                │
│                                              │
│ 💡 Why for you: You're working on your Nidra │
│ Protocol - this will deepen your understanding │
│ of why the practices help.                   │
│                                              │
│ [Start Lesson]                               │
└──────────────────────────────────────────────┘
```

Recommendation Reasons:

- Active project relevance
- Current imbalance relevance
- Seasonal relevance
- Constitution relevance
- Learning path progression
- Previously expressed interest

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                      │
│                                                                                      │
│                                                                                      │
```

```
│
┌──────────────
```

```
┌──────────────
```

```
                                        ▼
┌──────────────
```

```
│ Get user context:      │ ── Constitution, imbalances,
projects
│ • Prakriti             │
│ • Vikriti              │
│ • Active projects      │
│ • Season               │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Generate               │ ── AI-powered personalized
suggestions
│ recommendations        │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Claude generates       │ ── Contextual welcome message
│ Twin message           │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Render dashboard       │
│ with all sections      │
└────────────────────────┘
```

4.3 Data Flow: Taking a Lesson

```
User starts lesson
        │
        ▼
┌────────────────────────┐
│ Load lesson content    │ ── Sections, media, quizzes
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Get user context       │ ── For personalized sections
│ for personalization    │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Generate personalized  │ ── "For Your Constitution"
sections
│ context sections       │
```

```
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Track progress:          │ ── Sections viewed, time spent
│ • Section completion     │
│ • Time on lesson         │
└──────────────────────────┘
        │
        ▼
User completes section
        │
        ▼
┌──────────────────────────┐
│ If quiz section:         │
│ • Present question       │
│ • Score answer           │
│ • Show feedback          │
└──────────────────────────┘
        │
        ▼
User completes lesson
        │
        ▼
┌──────────────────────────┐
│ Calculate rewards:       │
│ • Base XP                │
│ • Quiz bonus             │
│ • Streak bonus           │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Update progress:         │
│ • Mark lesson complete   │
│ • Add XP                 │
│ • Check level up         │
│ • Update path progress   │
│ • Check achievements     │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Generate practical       │ ── Task suggestion for Today
section
│ application task         │
```

```
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Claude generates         │ ── Celebration, key takeaways
│ completion summary       │
└──────────────────────────┘
```

4.4 Data Flow: Learn in Context

```
User taps Ayurvedic term (anywhere in app)
        │
        ▼
┌──────────────────────────┐
│ Identify concept         │ ── Match term to concept ID
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Load concept summary     │ ── Quick definition, key points
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Get user context         │ ── Constitution, current state
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Generate personalized    │ ── "Your [Concept]" section
│ context                  │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Show quick concept       │ ── Modal/bottom sheet
│ card                     │
└──────────────────────────┘
        │
        ▼
User can:
├── Dismiss (return to original context)
├── [Learn More] → Open full concept page
└── [Related Lesson] → Start relevant lesson
```

4.5 Recommendation Algorithm

```
RECOMMENDATION ENGINE:
```

```
Inputs:
- User profile (Prakriti, Vikriti, imbalances)
- Active projects
- Current season
- Completed lessons
- Current learning path progress
- Time since last lesson

Scoring Factors:

1. PROJECT RELEVANCE (High weight)
   - User has Nidra Protocol → Sleep lessons +50
   - User has Agni Protocol → Digestion lessons +50

2. CONSTITUTION RELEVANCE (Medium weight)
   - Elevated Vata → Vata-related lessons +30
   - Elevated Pitta → Pitta-related lessons +30

3. SEASONAL RELEVANCE (Medium weight)
   - It's winter → Seasonal Living lessons +25
   - Approaching season change → Transition lessons +25

4. PATH PROGRESSION (Medium weight)
   - Next lesson in active path +35
   - Foundation incomplete → Foundation lessons +40

5. RECENCY (Low weight)
   - Haven't learned in 3+ days → Any lesson +10
   - Related to recent lesson → Continuation +15

6. LEVEL APPROPRIATENESS (Filter)
   - Only recommend unlocked content

Output:
- Ranked list of 3-5 recommended lessons
- "Why for you" explanation for each
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Learn Section

Claude serves as a teacher, guide, and explainer:

Capability	Description

Personalized Recommendations	Suggest lessons based on user's health journey
Contextual Explanations	Explain concepts in relation to user's constitution
Q&A	Answer questions during and after lessons
Personalized Content	Generate "For Your Constitution" sections
Practical Application	Suggest tasks based on learned concepts
Encouragement	Celebrate progress, motivate continued learning

5.2 System Prompt for Learn Section

```
You are the Ayurvedic Digital Twin for AyuAware, serving as a
knowledgeable teacher and learning guide. Your role is to
help users
build deep understanding of Ayurvedic principles, connecting
concepts
to their personal health journey.

## YOUR ROLE IN THIS SECTION

You are a teacher and guide. You explain complex Ayurvedic
concepts
in accessible ways, always connecting them to the user's
specific
constitution and health situation. You encourage learning
while
ensuring understanding is practical, not just theoretical.

## TEACHING PRINCIPLES

Personalize Everything:
- Connect concepts to user's Prakriti
- Relate to their current Vikriti/imbalances
- Link to their active projects
- Reference their health data where relevant

Balance Depth and Accessibility:
- Sanskrit terms with English explanations
- Traditional concepts with modern parallels
- Theory always connected to practice
- Progressive complexity (match their level)

Encourage Application:
- Every concept should have practical relevance
```

- Suggest specific actions based on learning
- Connect lessons to daily life
- Reinforce through health tracking

AYURVEDIC TEACHING CONTENT

Core Concepts:
- Tridosha theory (Vata, Pitta, Kapha)
- Prakriti (constitution) and Vikriti (current state)
- The 20 Gunas (qualities)
- Panchamahabhutas (5 elements)
- Like increases like / Opposites balance

Body Systems:
- Dhatus (7 tissues): Rasa, Rakta, Mamsa, Meda, Asthi, Majja, Shukra
- Srotas (channels): Various channel systems
- Malas (wastes): Purisha, Mutra, Sveda
- Agni (digestive/metabolic fire)

Health Markers:
- Agni states (Sama, Vishama, Tikshna, Manda)
- Ama (toxins) and its signs
- Ojas (vitality/immunity)
- Prana (life force)

Practices:
- Dinacharya (daily routine)
- Ritucharya (seasonal routine)
- Svasthavritta (self-care)
- Panchakarma (cleansing)
- Rasayana (rejuvenation)

COMMUNICATION APPROACH

When explaining concepts:
- Start with the Sanskrit term and translation
- Give clear, simple definition
- Explain relevance to the user specifically
- Provide practical application

When answering questions:
- Acknowledge the question
- Give direct answer first
- Add relevant context

```
- Connect to their situation

When celebrating progress:
- Be genuinely encouraging
- Highlight specific achievements
- Connect learning to health benefits
- Suggest next steps

## WHAT NOT TO DO

- Don't give overwhelming information at once
- Don't use jargon without explanation
- Don't teach without personalizing
- Don't leave concepts theoretical
- Don't ignore user's level (beginner vs advanced)
- Don't contradict established Ayurvedic principles
- Don't give medical advice (educational only)
```

5.3 Context Building for Learn Section

Every Claude call includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Prakriti: {prakriti_percentages}
- Primary constitution: {constitution_type}
- Current Vikriti: {vikriti_percentages}
- Current imbalances: {imbalance_list}
- Health scores: {Agni, Ojas, etc.}

## Learning Profile
- Current level: {level_number} ({level_name})
- XP: {current_xp} / {next_level_xp}
- Lessons completed: {count}
- Paths completed: {path_list}
- Active path: {path_name}
- Current lesson: {lesson_name} (if in progress)
- Learning streak: {days}

## Active Projects (if any)
- Project name: {project_name}
- Project focus: {focus_area}
- Related concepts: {concept_list}
```

```
## Current Context
- Current season: {season}
- Time of day: {time_period}
- Recent lessons: {last_3_lessons}

## Specific Request
- What user is asking/doing: {query_context}
```

5.4 Prompt Templates by Scenario

Dashboard Welcome:

```
Generate a welcome message for the Learn dashboard.

{CONTEXT_PACKAGE}

Generate a message that:

1. **Acknowledges their learning journey**
   - Current level/progress
   - Recent activity or streak

2. **Makes a specific recommendation**
   - Based on active projects, imbalances, or season
   - Explain WHY this lesson would help them

3. **Motivates continued learning**
   - Progress toward next level
   - Connection to health benefits

Tone: Encouraging, specific, teacher-like
Length: 3-4 sentences
```

Lesson Personalization:

```
Generate personalized content for this lesson section.

{CONTEXT_PACKAGE}

Lesson: {lesson_name}
Section: {section_name}
Section Content: {section_content}

Generate a "For Your Constitution" box that:

1. **Connects the concept to their Prakriti**
   - How this affects them specifically
   - What to watch for
```

```
2. **Relates to their current state**
   - Current Vikriti relevance
   - Active project connections

3. **Gives personalized guidance**
   - Specific application for them
   - What they should focus on

Length: 3-5 sentences
Tone: Informative, personal, practical
```

Concept Explanation (Q&A):

```
Explain this Ayurvedic concept to the user.

{CONTEXT_PACKAGE}

Concept: {concept_name}
User Question: {question}

Generate an explanation that:

1. **Defines the concept clearly**
   - Sanskrit term with meaning
   - Simple definition
   - Key characteristics

2. **Personalizes for them**
   - How it relates to their constitution
   - Current relevance

3. **Provides practical context**
   - How they encounter this in daily life
   - What to do with this knowledge

4. **Suggests further learning** (if appropriate)
   - Related concepts
   - Relevant lessons

Tone: Teacher explaining to interested student
Length: Appropriate to question complexity
```

Lesson Completion:

```
Generate a completion summary for this lesson.

{CONTEXT_PACKAGE}
```

```
Lesson: {lesson_name}
Time Spent: {duration}
Quiz Score: {score} / {total}
XP Earned: {xp}

Generate a summary that:

1. **Celebrates completion**
   - Acknowledge the achievement
   - Note quiz performance (if good)

2. **Summarizes key takeaways**
   - 3-4 main points from lesson
   - Personalized to their constitution

3. **Suggests practical application**
   - One specific action for today
   - Connected to lesson content
   - Relevant to their current state

4. **Points to next steps**
   - Next lesson in path
   - Related concepts to explore

Tone: Celebratory, reinforcing, forward-looking
```

Recommendation Generation:

```
Generate lesson recommendations for this user.

{CONTEXT_PACKAGE}

Available Lessons (not completed):
{available_lessons_list}

Generate 3 recommendations with:

For each recommendation:
1. **Lesson name**
2. **Duration and level**
3. **Why for you** (personalized reason)
   - Connect to projects, imbalances, season, or path
progress
   - Be specific about the benefit
```

```
Prioritize based on:
1. Active project relevance
2. Current imbalance relevance
3. Seasonal appropriateness
4. Path progression
5. Level appropriateness

Format as structured data for display.
```

Learn in Context:

```
Generate a quick explanation for this concept (Learn in
Context).

{CONTEXT_PACKAGE}

Concept: {concept_name}
Context: User tapped this term in {section_name}

Generate a quick card that:

1. **Brief definition** (1-2 sentences)
2. **Key qualities/characteristics**
3. **User's status** for this concept
   - Their type/level if applicable
   - Current relevance

Keep extremely concise - this is a quick reference card.
Length: 3-5 lines max
```

6. Data Models & State Management

6.1 Core Data Types

LearningPath:

```
LearningPath:
  - id: string
  - name: string
  - description: string
  - icon: string
  - category: 'beginner' | 'intermediate' | 'advanced' |
'expert'
  - requiredLevel: number
  - estimatedDuration: number (minutes)
  - lessons: PathLesson[]
  - prerequisites: string[] (path IDs)
```

```
  - unlocks: string[] (what completing this unlocks)

PathLesson:
  - lessonId: string
  - order: number
  - isRequired: boolean
```

Lesson:

```
Lesson:
  - id: string
  - pathId: string
  - name: string
  - description: string
  - duration: number (minutes)
  - difficulty: 'beginner' | 'intermediate' | 'advanced'
  - contentType: 'text' | 'video' | 'interactive' | 'mixed'
  - sections: LessonSection[]
  - quizzes: Quiz[]
  - keyTakeaways: string[]
  - relatedConcepts: string[]
  - practicalApplication: PracticalApplication
  - xpReward: number
  - tags: string[]

LessonSection:
  - id: string
  - order: number
  - title: string
  - contentType: 'text' | 'image' | 'video' | 'interactive' |
'personalized'
  - content: string | MediaContent | InteractiveContent
  - personalizationPrompt: string | null (for personalized
sections)

MediaContent:
  - type: 'image' | 'video'
  - url: string
  - caption: string | null
  - duration: number | null (for video)

InteractiveContent:
  - type: 'diagram' | 'slider' | 'tap_explore' | 'comparison'
  - data: any (interactive-specific data)

Quiz:
```

```
  - id: string
  - afterSection: number (section order)
  - question: string
  - type: 'multiple_choice' | 'true_false' | 'matching'
  - options: QuizOption[]
  - correctAnswer: string | string[]
  - explanation: string
  - xpBonus: number

QuizOption:
  - id: string
  - text: string

PracticalApplication:
  - description: string
  - suggestedTask: string
  - taskCategory: 'lifestyle' | 'nutrition' | 'practice' |
'observation'
```

Concept:

```
Concept:
  - id: string
  - name: string
  - sanskritName: string
  - sanskritScript: string
  - meaning: string
  - category: ConceptCategory
  - definition: string
  - detailedDescription: string
  - types: ConceptType[] | null
  - characteristics: string[]
  - doshaRelation: DoshaRelation | null
  - personalizable: boolean
  - personalizationPrompt: string | null
  - relatedConcepts: string[]
  - relatedLessons: string[]
  - tags: string[]

ConceptCategory:
  - 'core' | 'body_systems' | 'health_markers' | 'practices'
|
    'nutrition' | 'substances' | 'treatments'

ConceptType:
  - id: string
```

```
  - name: string
  - description: string
  - characteristics: string[]

DoshaRelation:
  - vata: string | null
  - pitta: string | null
  - kapha: string | null
```

UserProgress:

```
UserProgress:
  - userId: string
  - level: number
  - levelName: string
  - currentXP: number
  - totalXP: number
  - xpToNextLevel: number
  - completedLessons: CompletedLesson[]
  - completedPaths: string[]
  - activePath: string | null
  - currentLesson: LessonProgress | null
  - streak: StreakData
  - achievements: Achievement[]
  - conceptsLearned: string[]
  - totalTimeSpent: number (minutes)
  - quizAverageScore: number

CompletedLesson:
  - lessonId: string
  - completedAt: Date
  - timeSpent: number
  - quizScore: number | null
  - xpEarned: number

LessonProgress:
  - lessonId: string
  - startedAt: Date
  - lastAccessedAt: Date
  - currentSection: number
  - sectionsCompleted: number[]
  - quizResults: QuizResult[]
  - timeSpent: number

QuizResult:
  - quizId: string
```

```
  - answer: string | string[]
  - correct: boolean
  - answeredAt: Date

StreakData:
  - currentStreak: number
  - longestStreak: number
  - lastLearnDate: Date
  - streakStartDate: Date

Achievement:
  - id: string
  - name: string
  - description: string
  - icon: string
  - unlockedAt: Date
  - category: 'progress' | 'streak' | 'mastery' |
'exploration'
```

Recommendation:

```
Recommendation:
  - lessonId: string
  - lessonName: string
  - pathName: string
  - duration: number
  - difficulty: string
  - reason: string (personalized "why for you")
  - relevanceScore: number
  - relevanceType: 'project' | 'constitution' | 'seasonal' |
'path' | 'interest'
```

6.2 State Structure

```
LearnState:
  # Visit Context
  - isFirstVisit: boolean
  - lastVisit: Date | null

  # User Progress
  - progress: UserProgress | null
  - isLoadingProgress: boolean
  - progressError: Error | null

  # Paths
  - paths: LearningPath[]
  - pathStatuses: Record<string, PathStatus>
```

```
  - isLoadingPaths: boolean

  # Current Lesson
  - activeLesson: Lesson | null
  - lessonProgress: LessonProgress | null
  - personalizedSections: Record<string, string>
  - isLoadingLesson: boolean
  - currentSectionIndex: number
  - quizState: QuizState | null

  # Concept Library
  - concepts: Concept[]
  - conceptCategories: ConceptCategory[]
  - activeConcept: Concept | null
  - conceptPersonalization: string | null
  - isLoadingConcept: boolean

  # Recommendations
  - recommendations: Recommendation[]
  - isLoadingRecommendations: boolean

  # Search
  - searchQuery: string
  - searchResults: SearchResult[]

  # UI State
  - currentView: 'dashboard' | 'paths' | 'lesson' | 'concept'
| 'progress'
  - conceptModalOpen: boolean
  - conceptModalData: QuickConceptData | null
  - expandedPaths: string[]
  - scrollPosition: number

  # Chat
  - messages: LearnMessage[]
  - isTyping: boolean

PathStatus:
  - status: 'locked' | 'available' | 'active' | 'completed'
  - progress: number (0-100)
  - completedLessons: number
  - totalLessons: number
  - currentLesson: string | null

QuizState:
```

```
  - quizId: string
  - selectedAnswer: string | null
  - submitted: boolean
  - correct: boolean | null
  - showExplanation: boolean
```

6.3 State Management Approach

Zustand Stores:

```
Stores:

1. progressStore
   - User level and XP
   - Completed lessons/paths
   - Streaks
   - Achievements
   - Progress updates

2. contentStore
   - Paths and lessons
   - Concept library
   - Content caching
   - Search

3. lessonStore
   - Active lesson state
   - Section navigation
   - Quiz state
   - Personalized content
   - Progress tracking

4. recommendationStore
   - Recommendations
   - Recommendation generation
   - Personalization context

5. learnUIStore
   - View state
   - Modal state
   - Messages
   - Navigation
```

7. Component Requirements

7.1 Dashboard Components

LearningJourneyCard

User's progress overview:

- Level badge (number, name)
- Progress bar to next level
- Lessons/XP to next level
- Quick stats (lessons, time, streak)
- View All Progress link

ContinueLearningCard

Resume in-progress lesson:

- Lesson name and description
- Progress bar
- Time remaining
- Path context
- "Why this matters" personalization
- Continue button

LearningPathsSection

Path summaries:

- Completed paths (collapsed)
- Active path (highlighted)
- Recommended next path
- Locked paths with requirements

RecommendedLessonsSection

Personalized recommendations:

- 2-3 lesson cards
- Duration and difficulty
- "Why for you" explanation
- Start button

TopicsMasteredSection

Completed topics grid:

- Topic badges
- Tap to review
- Completion indicators

ConceptLibraryQuickAccess

Quick access to concepts:

- Popular concepts grid
- Browse All link
- Search prompt

7.2 Paths Components

PathCard

Learning path summary:

- Icon and name
- Description
- Lesson count and duration
- Status badge (Complete/Active/Locked)
- Progress bar (if active)
- Expand/collapse for lesson list
- Action button (Start/Continue/Review)

PathLessonItem

Lesson within path:

- Status indicator (✓, →, ○, 🔒)
- Lesson name
- Duration
- Current indicator if active

PathDetailView

Expanded path information:

- Full description
- All lessons with status
- Prerequisites
- What it unlocks
- Twin explanation of path value

7.3 Lesson Components

LessonHeader

Lesson navigation:

- Back button
- Lesson name
- Path context
- Progress indicator

LessonContentArea

Main content display:

- Text rendering (markdown)
- Image display
- Video player
- Interactive elements
- Personalized sections (highlighted)

PersonalizedContextBox

"For Your Constitution" sections:

- Distinct styling (highlighted)
- Constitution-specific content
- Generated by Claude

LessonProgressIndicator

Progress through lesson:

- Section X of Y
- Progress bar
- Previous/Next navigation

QuizComponent

Knowledge check:

- Question display
- Multiple choice options
- Submit button
- Correct/incorrect feedback
- Explanation display
- XP bonus indicator

LessonCompletionCard

Completion summary:

- Celebration message
- Time spent
- Quiz score
- XP earned
- Level progress
- Key takeaways
- Practical application
- Next lesson button

7.4 Concept Library Components

ConceptCategoryCard

Category browser:

- Category icon and name
- Concept count
- Expand to show concepts

ConceptListItem

Concept in list:

- Name (English and Sanskrit)
- Brief description
- Tap to view full

ConceptDetailView

Full concept page:

- Name (Sanskrit with script)
- Definition
- Personalized section ("Your [Concept]")
- Types/subtopics

- Characteristics
- Dosha relations
- Related concepts (links)
- Related lessons (links)

QuickConceptModal

Learn in Context card:

- Concept name
- Brief definition
- Key characteristics
- User's status
- Learn More / Dismiss buttons

ConceptSearchBar

Search concepts:

- Search input
- Filter by category
- A-Z browser

7.5 Progress Components

LevelProgressCard

Level detail:

- Current level badge
- XP progress bar
- What next level unlocks
- Lessons remaining

StatsGrid

Statistics display:

- Total lessons
- Total time
- Quiz average
- Current streak
- Paths complete
- Concepts learned

AchievementGrid

Achievements display:

- Unlocked achievements
- Locked achievements (grayed)
- Tap for details

LearningHistoryList

History timeline:

- Lessons completed by date
- Filter by month

- Tap to review

7.6 Animation Requirements

Animation	Duration	Easing	Trigger
Level up	1000ms	bounce	XP crosses threshold
XP gain	400ms	ease-out	Earn XP
Achievement unlock	800ms	bounce	Achievement triggered
Quiz correct	300ms	ease-out	Correct answer
Quiz incorrect	200ms	shake	Wrong answer
Lesson complete	600ms	ease-out	Finish lesson
Path complete	1000ms	bounce	Complete all lessons
Section transition	300ms	ease-out	Next/prev section
Concept modal	300ms	ease-out	Tap term
Progress bar fill	600ms	ease-out	Progress update

8. API Requirements

8.1 REST Endpoints

Get Learning Dashboard:

```
GET /api/v1/learn/dashboard

Response:
{
  "progress": {
    "level": 3,
    "levelName": "Apprentice",
    "currentXP": 450,
    "xpToNextLevel": 150,
    "totalLessonsCompleted": 12,
    "totalTimeSpent": 272,
    "streak": {
      "current": 5,
      "longest": 12
    }
  },
```

```
  "continueLesson": {
    "lessonId": "lesson_agni",
    "lessonName": "Understanding Agni",
    "pathName": "Nutrition & Digestion",
    "progress": 50,
    "timeRemaining": 12
  },
  "pathStatuses": [
    {
      "pathId": "path_foundation",
      "status": "completed",
      "progress": 100
    },
    {
      "pathId": "path_nutrition",
      "status": "active",
      "progress": 50,
      "currentLesson": "lesson_agni"
    },
    ...
  ],
  "recommendations": [
    {
      "lessonId": "lesson_sleep_doshas",
      "lessonName": "Sleep & the Doshas",
      "pathName": "Sleep & Recovery",
      "duration": 15,
      "difficulty": "intermediate",
      "reason": "You're working on your Nidra Protocol..."
    },
    ...
  ],
  "masteredTopics": ["doshas", "prakriti", "six_tastes", ...]
}
```

Get Learning Paths:

```
GET /api/v1/learn/paths

Response:
{
  "paths": [
    {
      "id": "path_foundation",
      "name": "Foundation Path",
```

```
      "description": "Essential concepts every student
needs",
      "icon": "📘",
      "category": "beginner",
      "requiredLevel": 1,
      "estimatedDuration": 150,
      "lessonCount": 7,
      "status": "completed",
      "progress": 100,
      "lessons": [
        {
          "lessonId": "lesson_what_is_ayurveda",
          "name": "What is Ayurveda?",
          "duration": 12,
          "status": "completed"
        },
        ...
      ]
    },
    ...
  ]
}
```


Get Lesson Content:

```
GET /api/v1/learn/lessons/:lessonId

Response:
{
  "lesson": {
    "id": "lesson_agni",
    "name": "Understanding Agni",
    "description": "Your digestive fire determines your
health",
    "pathId": "path_nutrition",
    "pathName": "Nutrition & Digestion",
    "duration": 15,
    "difficulty": "intermediate",
    "sections": [
      {
        "id": "section_1",
        "order": 1,
        "title": "What is Agni?",
        "contentType": "text",
        "content": "Agni is the Sanskrit word for 'fire'..."
      },
```

```
      {
        "id": "section_2",
        "order": 2,
        "title": "For Your Constitution",
        "contentType": "personalized",
        "personalizationPrompt": "Explain Agni relevance..."
      },
      ...
    ],
    "quizzes": [
      {
        "id": "quiz_1",
        "afterSection": 3,
        "question": "Which Agni type is associated with
Vata?",
        "type": "multiple_choice",
        "options": [
          { "id": "a", "text": "Sama Agni" },
          { "id": "b", "text": "Tikshna Agni" },
          { "id": "c", "text": "Vishama Agni" },
          { "id": "d", "text": "Manda Agni" }
        ],
        "correctAnswer": "c",
        "explanation": "Vishama Agni is associated with
Vata..."
      }
    ],
    "keyTakeaways": [
      "Agni is your digestive & metabolic fire",
      ...
    ],
    "practicalApplication": {
      "description": "Apply your Agni knowledge",
      "suggestedTask": "Eat meals at consistent times today",
      "taskCategory": "lifestyle"
    },
    "xpReward": 50,
    "relatedConcepts": ["agni", "ama", "dhatus"]
  },
  "userProgress": {
    "started": true,
    "currentSection": 2,
    "sectionsCompleted": [1],
    "timeSpent": 8
  }
```

```
}
```

Update Lesson Progress:

```
POST /api/v1/learn/lessons/:lessonId/progress

Request:
{
  "sectionCompleted": 3,
  "timeSpent": 4
}

Response:
{
  "success": true,
  "progress": {
    "sectionsCompleted": [1, 2, 3],
    "totalSections": 5,
    "percentComplete": 60
  }
}
```

Submit Quiz Answer:

```
POST /api/v1/learn/lessons/:lessonId/quiz/:quizId

Request:
{
  "answer": "c"
}

Response:
{
  "correct": true,
  "explanation": "Vishama Agni is associated with Vata...",
  "xpBonus": 10
}
```

Complete Lesson:

```
POST /api/v1/learn/lessons/:lessonId/complete

Request:
{
  "timeSpent": 18,
  "quizResults": [
    { "quizId": "quiz_1", "correct": true },
    { "quizId": "quiz_2", "correct": true }
  ]
```

```
}

Response:
{
  "success": true,
  "xpEarned": {
    "base": 50,
    "quizBonus": 10,
    "streakBonus": 5,
    "total": 65
  },
  "progress": {
    "previousLevel": 3,
    "newLevel": 3,
    "currentXP": 515,
    "xpToNextLevel": 85,
    "leveledUp": false
  },
  "pathProgress": {
    "pathId": "path_nutrition",
    "completedLessons": 5,
    "totalLessons": 8,
    "percentComplete": 62
  },
  "achievementsUnlocked": [],
  "streakUpdated": {
    "current": 6,
    "continued": true
  },
  "practicalTask": {
    "task": "Eat meals at consistent times today",
    "addedToToday": true
  }
}
```

Get Concept:

```
GET /api/v1/learn/concepts/:conceptId

Response:
{
  "concept": {
    "id": "agni",
    "name": "Agni",
    "sanskritName": "Agni",
    "sanskritScript": "अग्नि",
```

```
    "meaning": "Fire, Digestive Fire",
    "category": "health_markers",
    "definition": "Agni is the transformative principle...",
    "detailedDescription": "In Ayurveda, Agni represents...",
    "types": [
      {
        "id": "jatharagni",
        "name": "Jatharagni",
        "description": "Main digestive fire..."
      },
      ...
    ],
    "characteristics": ["Transformation", "Metabolism",
"Digestion"],
    "doshaRelation": {
      "vata": "Vishama (variable) Agni",
      "pitta": "Tikshna (sharp) Agni",
      "kapha": "Manda (slow) Agni"
    },
    "relatedConcepts": ["ama", "dhatus", "ojas"],
    "relatedLessons": ["lesson_agni",
"lesson_digestion_signs"]
  },
  "personalization": {
    "userAgniType": "Vishama",
    "relevance": "Your Agni type is likely Vishama..."
  }
}
```


Get Concept Library:

```
GET /api/v1/learn/concepts?category=health_markers

Response:
{
  "categories": [
    {
      "id": "health_markers",
      "name": "Health Markers",
      "concepts": [
        {
          "id": "agni",
          "name": "Agni",
          "sanskritName": "Agni",
          "briefDescription": "Digestive and metabolic fire"
        },
```

```
        ...
      ]
    },
    ...
  ]
}
```

Get Quick Concept (Learn in Context):

```
GET /api/v1/learn/concepts/:conceptId/quick

Response:
{
  "concept": {
    "id": "vata",
    "name": "Vata",
    "briefDefinition": "The dosha of movement and change...",
    "keyQualities": ["Light", "Dry", "Cold", "Mobile",
"Rough"],
    "userStatus": "Currently elevated (40% vs 25% baseline)"
  }
}
```

Get Progress Detail:

```
GET /api/v1/learn/progress

Response:
{
  "level": {
    "current": 3,
    "name": "Apprentice",
    "currentXP": 450,
    "totalXP": 450,
    "xpToNextLevel": 150,
    "nextLevelUnlocks": [
      "Advanced Practices Path",
      "Dosha Deep Dive Path",
      "Protocol customization"
    ]
  },
  "statistics": {
    "totalLessons": 12,
    "totalTime": 272,
    "quizAverage": 84,
    "currentStreak": 5,
    "longestStreak": 12,
    "pathsCompleted": 1,
```

```
    "conceptsLearned": 24
  },
  "achievements": [
    {
      "id": "first_lesson",
      "name": "First Steps",
      "description": "Complete your first lesson",
      "icon": "🌱",
      "unlockedAt": "2026-01-15"
    },
    ...
  ],
  "history": [
    {
      "date": "2026-01-30",
      "lessons": [
        { "lessonId": "lesson_agni", "progress": 50 }
      ]
    },
    ...
  ]
}
```

8.2 WebSocket Events (Optional)

```
WebSocket Events:

learn.level_up
{
  "previousLevel": 3,
  "newLevel": 4,
  "newLevelName": "Practitioner",
  "unlockedContent": ["Advanced Practices Path", ...]
}

learn.achievement_unlocked
{
  "achievementId": "quiz_master",
  "name": "Quiz Master",
  "description": "Score 100% on 10 quizzes"
}

learn.streak_milestone
{
  "streakDays": 7,
```

```
  "message": "7 day learning streak! 🔥"
}
```

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Section visits per week	> 3	Users visiting Learn section
Lessons started per week	> 2	Lessons begun by users
Time in section	> 10 min/visit	Average session duration
Return rate	> 50%	Users returning within 7 days

9.2 Completion Metrics

Metric	Target	Measurement
Lesson completion rate	> 70%	Lessons finished vs. started
Path completion rate	> 40%	Paths finished vs. started
Foundation completion	> 60%	Users who complete Foundation path
Quiz completion rate	> 80%	Quizzes answered vs. encountered

9.3 Learning Quality Metrics

Metric	Target	Measurement
Quiz average score	> 75%	Average quiz performance
Concept retention	> 60%	Ability to answer later quizzes
Practical application	> 30%	Users who complete suggested tasks
Level progression	> 50%	Users who reach Level 3+

9.4 Feature Usage Metrics

Metric	Target	Measurement
Path exploration	> 40%	Users who view multiple paths

Concept Library usage	> 25%	Users who browse concepts
Learn in Context usage	> 20%	Users who tap terms elsewhere
Recommendation acceptance	> 30%	Recommended lessons started

9.5 Retention Metrics

Metric	Target	Measurement
Learning streak average	> 3 days	Average streak length
Weekly active learners	> 40%	Users who learn each week
30-day retention	> 30%	Users still learning after 30 days
Path restart rate	< 20%	Users who restart abandoned paths

9.6 Technical Metrics

Metric	Target	Measurement
Dashboard load time	< 1s	Time to display dashboard
Lesson load time	< 2s	Time to load lesson content
Quiz response time	< 500ms	Time to confirm quiz answer
Personalization time	< 2s	Time to generate personalized sections
Error rate	< 1%	Failed operations

10. Implementation Considerations

10.1 Content Management

Lesson Content:

- Store lesson content as structured data (not hardcoded)
- Support markdown for text sections
- Media stored in CDN
- Version control for content updates
- Translation-ready structure

Concept Library:

- Comprehensive Ayurvedic concept database
- Regular content updates
- Cross-referencing system
- Search indexing

Personalization Templates:

- Prompts stored with lessons/concepts
- Cacheable personalized content
- Fallback to generic if personalization fails

10.2 Progress Persistence

Local Storage:

- Cache lesson progress
- Queue XP updates
- Store quiz answers
- Maintain streak locally

Server Sync:

- Sync progress on lesson complete
- Periodic background sync
- Conflict resolution (server wins)
- Offline progress preserved

10.3 Learn in Context (App-Wide)

Implementation:

- Wrap Ayurvedic terms in tappable components
- Global concept lookup service
- Quick concept modal accessible everywhere
- Deep links to full concept pages

Term Detection:

- Maintain list of Ayurvedic terms
- Pattern matching in text
- Manual markup in UI components

10.4 Integration Points

Today Section:

- Practical application tasks
- Learning reminders
- Streak notifications

Projects Section:

- Recommend lessons for active projects
- Link learning to protocol understanding
- Concept references in protocols

Health Section:

- Explain health markers
- Link metrics to concepts
- Educational context for scores

All Sections:

- Learn in Context (tap terms)
- Concept quick cards
- Deep links to lessons

10.5 Notification System

Learning Reminders:

- Daily learning reminder (configurable)
- Streak at risk notification
- New content available

Progress Notifications:

- Level up celebration
- Achievement unlocked
- Path completed

Personalized Prompts:

- "This would help your Nidra Protocol..."
- "Seasonal learning suggestion..."

10.6 Offline Handling

When Offline:

- View cached lessons
- Continue in-progress lessons
- Take quizzes (results queued)
- Browse cached concepts
- View progress

When Back Online:

- Sync queued progress
- Update XP and level
- Check achievements
- Refresh recommendations

10.7 Accessibility Requirements

Screen Reader:

- Lesson content accessible
- Quiz options announced
- Progress communicated
- Navigation clear

Motor Accessibility:

- Large tap targets
- Swipe navigation
- No time-sensitive interactions

Visual Accessibility:

- WCAG AA contrast
- Readable font sizes
- Color not sole indicator
- Support Dynamic Type

10.8 Analytics Events

Event	Properties	Purpose
`learn_opened`	source	Track entry
`lesson_started`	lesson_id, path_id	Engagement
`lesson_completed`	lesson_id, time_spent, quiz_score	Completion
`lesson_abandoned`	lesson_id, section_reached	Drop-off
`quiz_answered`	quiz_id, correct	Learning quality
`path_started`	path_id	Path engagement
`path_completed`	path_id, total_time	Path completion
`concept_viewed`	concept_id, source	Concept interest
`concept_quick_viewed`	concept_id, context	Learn in Context
`level_up`	new_level	Progression
`achievement_unlocked`	achievement_id	Gamification
`recommendation_accepted`	lesson_id, reason_type	Recommendation quality
`search_performed`	query, results_count	Search usage

10.9 Testing Priorities

Unit Tests:

- XP calculation
- Level progression
- Quiz scoring
- Streak calculation
- Progress tracking

Integration Tests:

- Lesson completion flow
- Quiz submission and feedback
- Progress sync
- Recommendation generation
- Learn in Context modal

E2E Tests:

- Complete Foundation Path
- Level up experience
- Path unlock after level up
- Learn in Context from Health section

Manual QA:

- Lesson content accuracy
- Personalization relevance
- Quiz feedback clarity
- Progress display accuracy
- Cross-section linking

Appendix A: Learning Path Curriculum

Foundation Path (Required)

Lesson	Duration	Key Concepts	Quiz Topics
What is Ayurveda?	12 min	History, Vedas, principles	Origins, core principles
The Three Doshas	15 min	Vata, Pitta, Kapha	Dosha identification
Discovering Your Prakriti	18 min	Constitution, assessment	Prakriti types
Understanding Vikriti	15 min	Imbalance, current state	Prakriti vs Vikriti
The 20 Gunas	12 min	Qualities, opposites	Guna pairs
Like Increases Like	10 min	Similarity principle	Application examples
Your Digital Twin & Ayurveda	10 min	App philosophy	Feature understanding

Nutrition & Digestion Path

Lesson	Duration	Key Concepts	Quiz Topics
The Six Tastes	15 min	Rasa, dosha effects	Taste-dosha relations
Food Combining	12 min	Compatible/ incompatible	Combining rules
Eating for Your Dosha	18 min	Constitutional diet	Food recommendations
Understanding Agni	15 min	Digestive fire types	Agni states
Signs of Healthy Digestion	12 min	Assessment markers	Health indicators
Ama: Toxins	15 min	Toxin formation, signs	Ama identification
Seasonal Eating	12 min	Ritucharya nutrition	Seasonal foods
Fasting & Cleansing	15 min	Safe approaches	Fasting guidelines

Appendix B: Concept Library Structure

Category: Core Concepts

Concept	Sanskrit	Related Lessons
Vata	वात	Three Doshas, Prakriti
Pitta	पित्त	Three Doshas, Prakriti
Kapha	कफ	Three Doshas, Prakriti

Prakriti	प्रकृति	Discovering Your Prakriti
Vikriti	विकृति	Understanding Vikriti
Gunas	गुण	The 20 Gunas
Panchamahabhutas	पञ्चमहाभूत	What is Ayurveda

Category: Health Markers

Concept	Sanskrit	Related Lessons
Agni	अग्नि	Understanding Agni
Ama	आम	Ama: Toxins
Ojas	ओजस	Building Ojas
Prana	प्राण	Mind-Body Connection
Tejas	तेजस	Mental Qualities

Category: Body Systems

Concept	Sanskrit	Related Lessons
Dhatus	धातु	Advanced: Tissue System
Srotas	स्रोतस	Advanced: Channels
Malas	मल	Signs of Healthy Digestion

Appendix C: Achievement System

Progress Achievements

Achievement	Icon	Criteria
First Steps	🌱	Complete first lesson
Foundation Complete	📘	Complete Foundation Path
Path Master	🛤️	Complete any 3 paths
Scholar	🎓	Reach Level 6
Master	👑	Reach Level 7

Streak Achievements

Achievement	Icon	Criteria
Consistent	🔥	7-day streak
Dedicated	💪	30-day streak
Committed	🏆	100-day streak

Mastery Achievements

Achievement	Icon	Criteria
Quiz Master	🎯	Score 100% on 10 quizzes
Perfect Path	⭐	Complete a path with all quizzes 100%
Concept Explorer	📚	View 30+ concepts

Exploration Achievements

Achievement	Icon	Criteria
Curious Mind	🔍	Use Learn in Context 10 times
Deep Diver	🌊	Complete all advanced paths
Completionist	✨	Complete all lessons

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